A Dialog on the Quantum Group Fourier Transform

Course Paper for PHYS 3450: Intro. to Quantum Computing

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Introduction

A "generic" quantum algorithm involving an oracle typically takes the form of a careful preparation of a superposition of states to pass into said oracle, followed by a sequence of measurements and gate operations which induce a sufficient amount of quantum interference to collapse and rotate the resulting superposition down to a useful state.

The *Quantum Group Fourier Transform* (QGFT) is truly delightful piece of Representation Theory which facilitates the construction of those *opening* and *closing* steps of quantum interference without any guesswork: utilizing the group-theoretic structure of the promised oracle to inform the QGFT gates applied. The purpose of this introductory note is to derive, develop, and build a robust intuition for the QGFT while relating it to instances of the classic quantum algorithms seen in PHYS 3450, supported by MATH 3500-level introductory abstract algebra.

More explicitly, we will begin our dialog with two sections of a more abstract treatment of the machinery behind the QGFT and then accompany this with two sections containing concrete algorithmic applications as furnished by the closely related and well-studied *Hidden Subgroup Problem*: with the goal of bridging representation theory with quantum circuitry in a (hopefully) intuitive way.

Some Representation Theory & Motivation

For our purposes, a representation ρ with dimension d of a group G is a homomorphism between G and $\mathrm{GL}_d(\mathbb{C})$: with the intution being that since $\rho(g)\rho(h) = \rho(gh)$ for all $g, h \in G$, we can reason about the abstract group action of G simply by multiplying the relevant $d \times d$ matrices $\rho(g)$ and $\rho(h)$. The representation *character* $\chi_\rho(g) : G \to \mathbb{C}$ is then defined accordingly by the trace $\mathrm{tr}(\rho(g))$, with the intuition being that since $\mathrm{tr}(PAP^{-1}) = \mathrm{tr}(A)$ is invariant under conjugation of matrices (one can easily check this formula explicitly), the character $\chi_\rho(hgh^{-1}) = \mathrm{tr}(\rho(h)\rho(g)\rho(h)^{-1}) = \chi_\rho(g)$ ends up being invariant under conjugation within G. As a side effect of the conjugation invariance of tr, χ_ρ does not depend on our choice of basis used to realize the matrix ρ.

For example, we know that multiplying 2×2 rotation matrices $R_\theta = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$ together corresponds to *compounding* the angle of rotation applied: namely we have the *homomorphism* rule $R_\theta R_\phi = R_{\theta+\phi}$ which tells us that the symmetry group of a circle S^1 can be elegantly encoded in the space of 2×2 invertible matrices. In the same, way $\mathbb{Z}_n$ has a nice 2×2 representation of the form $\rho(k) = \begin{bmatrix} \cos(2\pi k/n) & -\sin(2\pi k/n) \\ \sin(2\pi k/n) & \cos(2\pi k/n) \end{bmatrix}$ where given that such $2\pi/n$ counter-clockwise rotations eventually loop back around with period n, perfectly encodes the fact that the generator 1 will return periodically back to $\underbrace{1 + 1 + \ldots 1}_{n \text{ times}} = 0$. The same group also has a $d = 1$ dimensional representation: $\rho(k) = [e^{2\pi ik/n}]$ since clearly the complex exponential already constitutes this $2\pi/n$ rotation. On the other hand, we can create a bulkier $d = n$ representation like $\rho(k) = \begin{bmatrix} 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \end{bmatrix}^k$ which corresponds to executing k times the cyclic shift of the basis vectors $e_1, \ldots e_n$ sending $e_1 \mapsto e_2, \ldots e_n \mapsto e_1$: a cyclic shift which is clearly in correspondence with the periodic structure of $\mathbb{Z}_n$.

The theory of ρ and χ is equally vast as it is beautiful, but for our purposes we will only need the following notions from elementary Representation Theory (specifically borrowed from pages 870 through 876 in [1]):

- **Dual Group:** To describe the collection of all "meaningful" representations of G we write $\widehat{G}$ for the set of representations of G that satisfy all of:
  - *Unitary:* It is helpful to enforce $\rho(g)^\dagger \rho(g) = I$ for all g.
  - *Inequivalent:* Two linear transformations are spectrally identical when they differ by a similarity transform. For any $\rho_1 \neq \rho_2 \in \widehat{G}$, there should not exist a similarity transform $\rho_2(g) = T\rho_1(g)T^{-1}$ sending one to the other for all g.
  - *Irreducible:* ρ should not be composed of sub-representations. There should not exist a subspace $W \subseteq \mathbb{C}^d$ that is $\rho(g)$-invariant for all g, since if so ρ could simply be split into two smaller, simpler linear operators acting on W and its complement.
- **Character Space:** As ρ ranges over $\widehat{G}$, the set of χ_ρ forms an *orthonormal basis* for the "character space": specifically they are linearly independent, ie if $\sum_{\rho \in \widehat{G}} c_\rho \chi_\rho(g) = 0$ for all $g \in G$, then such a linear combination is trivial with $c_\rho = 0$ and they are orthonormal with respect to the Hermitian inner product $\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi}_1(g)\chi_2(g)$.
- **Irreducibility Criterion:** A representation ρ is irreducible if and only if $\langle \chi_\rho, \chi_\rho \rangle = 1$. In the above, we've written the "if" part.
- **Inequivalency Criterion:** Two representations ρ and ρ' are inequivalent if and only if $\langle \chi_\rho, \chi_{\rho'} \rangle \neq 0$.
- **Column Orthogonality:** Not only are the χ_ρ's orthogonal w.r.t a sum over G ("row orthogonality"), they are also orthogonal w.r.t a sum over $\widehat{G}$:

$$\sum_{\rho \in \widehat{G}} \overline{\chi_\rho(g)} \chi_\rho(h) = \begin{cases} 0 & g \nsim h \\ \frac{|G|}{|\mathrm{Cl}(g)|} & g \sim h \end{cases} \tag{1}$$

  where here $\sim$ stands for the relation of g and h being members of the same *conjugacy class*, and $|\mathrm{Cl}(g)|$ provides the number of elements in this conjugacy class. This result should be at least plausible given that χ_ρ is invariant under conjugation, meaning if g and h lie in the same conjugacy class, we'll have $\chi_\rho(g) = \chi_\rho(h)$.
- **Regular Representation:** For every finite group G, there exists a ρ, not necessarily irreducible, whose corresponding character χ_{reg} is such that $\chi_{\text{reg}}(g) = |G|\delta_{g=e}$, ie it vanishes on all inputs other than the identity, where it returns $|G|$.

This is a list for our reference: no need to absorb them all at once. However, what can be seen from this crash-course already is how powerful a mathematical tool the character χ is: acting as a detector for a remarkable array of important properties concerning the underlying representation ρ.

Exercise

Three representations for the group $\mathbb{Z}_n$ were given above: determine the characters $\chi_\rho(k)$ explicitly for each one. One of them happens to be χ_{reg}: which is it? Which of them are unitary? Find a $\rho(g)$-invariant subspace of the $n \times n$ representation: what does this tell you about irreducibility? Finally, for any representation ρ, prove that $\chi_\rho(e) = d_\rho$.

The main definition of this note is the following, which we will shortly unpack:

The *Quantum Group Fourier Transform* (QGFT)[1] corresponding to a finite group $G = \{g_0, \dots g_{n-1}\}$, denoted $\mathcal{F}_G$, is a unitary operator from the "time" domain $|G\rangle$, namely the Hilbert space generated by the computational basis vectors $|g_0\rangle, \dots |g_{n-1}\rangle$, to the "frequency domain" $\left|\widehat{G}\right\rangle$: namely the space generated by $|\rho_{i,j}\rangle$ as the representation ρ, which is a $d_\rho \times d_\rho$ matrix, ranges over $\widehat{G}$, and as $1 \le i, j \le d_\rho$ range over all d_ρ^2 entries in that matrix. In total, the definition works out to be:

Quantum Group Fourier Transform

Let G be a finite group and $\widehat{G}$, as above, its dual. Define:

$$\mathcal{F}_G |g\rangle = \sum_{\rho \in \widehat{G}} \sqrt{\frac{d_\rho}{|G|}} \sum_{1 \le i,j \le d_\rho} [\rho(g)]_{i,j} |\rho_{i,j}\rangle \tag{2}$$

Then, $\mathcal{F}_G$ is a unitary operator ie $\mathcal{F}_G^\dagger \mathcal{F}_G = \mathrm{id}_{|G\rangle}$.

Not only is it not obvious why the above combination of entries of representations should encode *anything* useful about the periodic structure of the group G, it's not even clear that $\mathcal{F}_G$ maps between two spaces of the same dimension! The solution, as well as the motivating reason for why $\left|\widehat{G}\right\rangle$ must range over all d_ρ^2 entries in a representation ρ rather than just ranging over the ρ's, is the following classic result of representation theory:

$$|G| = \sum_{\rho \in \widehat{G}} d_\rho^2 \tag{3}$$

Proof: Recall that the regular representation character χ_{reg} exists and can be expanded in the orthonormal basis of characters, meaning $\chi_{\text{reg}} = \sum_{\rho \in \widehat{G}} \langle \chi_{\text{reg}}, \chi_\rho \rangle \chi_\rho$. By definition, note that $\langle \chi_{\text{reg}}, \chi_\rho \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_{\text{reg}}(g) \overline{\chi_\rho(g)}$ and since χ_{reg} vanishes for every $g \in G$ except for at the identity e, this sum reduces down to just $\frac{1}{|G|} \chi_{\text{reg}}(e) \overline{\chi_\rho(e)} = \frac{1}{|G|} |G| \overline{d_\rho} = d_\rho$. So, we have $\chi_{\text{reg}}(g) = \sum_{\rho \in \widehat{G}} d_\rho \chi_\rho(g)$ which tells us two important facts: at $g = e$ we receive the desired $|G| = \sum_{\rho \in \widehat{G}} d_\rho \chi_\rho(e) = \sum_\rho d_\rho^2$ equality of dimensions, and for all other g, the sum $\sum_{\rho \in \widehat{G}} d_\rho \chi_\rho(g) = 0$ completely vanishes.

We'll use this vanishing / orthogonality relation to quickly start simplifying the inner product between $\mathcal{F}|g\rangle$ and $\mathcal{F}|g'\rangle$. First, since the $|\rho_{i,j}\rangle$'s form an orthonormal basis, we immediately collapse down to a single sum:

$$(\mathcal{F}|g\rangle)^\dagger (\mathcal{F}|g'\rangle) = \frac{1}{|G|} \sum_{\rho \in \widehat{G}} \sum_{\rho' \in \widehat{G}} \sqrt{d_\rho d_{\rho'}} \sum_{i,j}^{d_\rho} \sum_{i',j'}^{d_{\rho'}} \overline{[\rho(g)]_{i,j}} [\rho'(g)]_{i,j} \underbrace{\langle \rho_{i,j} | \rho'_{i,j} \rangle}_{\delta_{\rho=\rho', i=i', j=j'}} = \frac{1}{|G|} \sum_{\rho \in \widehat{G}} \sqrt{d_\rho^2} \sum_{i,j}^{d_\rho} \overline{[\rho(g)]_{i,j}} [\rho(g')]_{i,j}$$

Now, this innermost sum over i, j looks a lot like the inner product of two matrices $\mathrm{tr}(A^\top B) = \sum_{i,j} A_{i,j} B_{i,j}$, and indeed since the complex conjugate $\overline{\rho(g)}$ is just the conjugate transpose $(\rho(g)^\dagger)^\top$ in disguise, we have exactly $\sum_{i,j}^{d_\rho} [(\rho(g)^\dagger)^\top]_{i,j} [\rho(g')]_{i,j} = \mathrm{tr}(\rho(g)^\dagger \rho(g'))$. Even better, since each $\rho \in \widehat{G}$ is unitary by assumption, this trace can be reduced further down to $\mathrm{tr}(\rho^{-1}(g)\rho(g')) = \mathrm{tr}(\rho(g^{-1}g')) = \chi_\rho(g^{-1}g')$ and so finally by the orthogonality relation above:

$$(\mathcal{F}|g\rangle)^\dagger (\mathcal{F}|g'\rangle) = \frac{1}{|G|} \sum_{\rho \in \widehat{G}} d_\rho \chi_\rho(g^{-1}g') = \delta_{g^{-1}g'=e} = \delta_{g=g'} \tag{4}$$

And so we have established that the Quantum Group Fourier Transform is a unitary operator! A beautiful correspondence where representation theory and quantum computation collide.

[1] Funnily enough, the study of "quantum groups" and their representations is an important piece of modern mathematics, but one which I believe is entirely unrelated to the "quantum" implementation of "group Fourier transforms" that we desire here.

But beyond the nice unitarity properties of the QGFT, the reason why representations make any appearance at all in the theory of quantum algorithms is because they allow for the extraction of a *subgroup state*, namely a superposition of the form $|H\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |h\rangle$ for some subgroup $H \leq G$, from an arbitrary *coset state*, namely $|gH\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |gh\rangle$.

Intuitively, given a coset $gH = \{gh_0, \ldots gh_{m-1}\}$ without knowing g, it is not a trivial matter to deduce what g is or what H is: one could try looking at the ratios $(gh_i)^{-1}(gh_j) = h_i^{-1}h_j$ for example to eliminate its effect, but trying to guess the subgroup structure from these $\binom{m}{2}$ possible pairings requires some creativity.

Amazingly, a fairly small lemma from representation theory affords us a quite powerful tool for *measuring* $|H\rangle$ on a quantum computer [2]. That lemma is;

Projection Operators

Let $\rho \in \widehat{G}$ be a representation and let $H \leq G$ be an arbitrary subgroup. Let $V_\rho^H \subseteq \mathbb{C}^{d_\rho}$ be the subspace containing those vectors $v = \rho(h)v$ which are fixed for each $h \in H$, namely $V_\rho^H = \bigcap_{h \in H} \ker(\rho(h) - I)$. Then,

$$\Pi_\rho^H = \frac{1}{|H|} \sum_{h \in H} \rho(h) \tag{5}$$

Constitutes a Hermitian *projection* matrix onto the subspace V_ρ^H.

Why is this helpful in our quest to learn H from gH? Well consider what occurs when $\mathcal{F}_G$ acts on the coset state $|gH\rangle$:

$$\mathcal{F}_G(|gH\rangle) = \sqrt{|H|} \sum_{\rho \in \widehat{G}} \sqrt{\frac{d_\rho}{|G|}} \sum_{1 \leq i,j \leq d_\rho} \frac{1}{|H|} \sum_{h \in H} [\underbrace{\rho(gh)}_{\rho(g)\rho(h)}]_{i,j} |\rho_{i,j}\rangle = \sum_{\rho \in \widehat{G}} \sqrt{\frac{d_\rho |H|}{|G|}} \sum_{1 \leq i,j \leq d_\rho} [\rho(g)\Pi_\rho^H]_{i,j} |\rho_{i,j}\rangle$$

What's fascinating is that the probability of measuring $|\rho_{i,j}\rangle$ for *some* i, j (meaning we look only to the name of the representation ρ and not at which particular entry we have sampled) is proportional to $d_\rho \times \sum_{i,j} |\rho(g)\Pi_\rho^H]|_{i,j}^2$ by the Born rule. Using the same manipulations with trace and † as we used in the previous theorem, this is to say:

$$P(|\rho\rangle) \propto d_\rho \sum_{1 \leq i,j \leq d_\rho} [\Pi_\rho^{H\dagger} \rho(g)^\dagger]_{j,i} [\rho(g)\Pi_\rho^H]_{i,j} = d_\rho \mathrm{tr}\left(\Pi_\rho^{H\dagger} \underbrace{\rho(g)^\dagger \rho(g)}_{I} \Pi_\rho^H\right) = d_\rho \mathrm{tr}(\Pi_\rho^H) = d_\rho \dim V_\rho^H \tag{6}$$

Where we've used the fact that Π_ρ^H is both Hermitian, that it squares to itself, and a nice lemma from linear algebra that if Π is a projection matrix onto a subspace V, then $\mathrm{tr}(\Pi) = \dim V$, which follows from the fact that a basis $v_1, \ldots v_{\dim V}$ for V, when completed to a basis for the entire vectorspace, diagonalizes Π into a $\dim V \times \dim V$ identity block surrounded by 0s. In any case, what matters is that a key cancellation has occurred: the appearance of g has vanished from the probability of observing $|\rho\rangle$ and depends only on the underlying subgroup H we wish to study!

All the way down at 1 we will compute some explicit examples of these probability distributions over the representations ρ, but the key takeaway from 6 is that upon *measuring* the output register form a QGFT $\mathcal{F}_G$, we will never read out a state $|\rho_{i,j}\rangle$ such that V_ρ^H is 0-dimensional: we end up *learning* a constraint that H must satisfy with respect to ρ. It sounds abstract, but we'll see it repeatedly in : "a measurement of $\mathcal{F}_G$ gives a constraint H must satisfy".

Exercise

To take advantage of the "parallel" nature of a quantum oracle query, we almost always start quantum algorithms with a uniform superposition over the pure states in the system. How would you make $\frac{1}{\sqrt{|G|}}\sum_{g\in G}|g\rangle$?

Second, see if you can prove 5 by noticing that $\rho(h')\Pi_\rho^H = \Pi_\rho^H$, as obtained by the fact that $\sum_{h\in H}\rho(h')\rho(h) = \sum_{h'h\in H}\rho(h) = \sum_{h\in H}\rho(h)$ simply gets shuffled around. If you get stuck consider peeking at [2] page 22.

Finally, compute what Π_ρ^G is for the entire group G, recalling that ρ is irreducible and hence has no non-trivial invariant subspaces: what must V_ρ^G be then? Check your answer against our explicit calculations in 1: do they match?

Developing the Abelian QGFT

It needs no explanation that the *Discrete Fourier Transform* is one of, if not the, most important and influential coordinate transformations in both the applied sciences and pure mathematics: the powerful and versatile mirror which views a spatial or temporal signal $x(1), \dots x(n)$ as a sum of pure, oscillatory harmonics $e^{-\frac{2\pi i}{n}kx}$ for $0 \le k \le n-1$. For the purposes of quantum computation, the "signal" x is replaced with computational basis states $|0\rangle, \dots |n-1\rangle$, but the underlying formula and its purpose remains the same:

$$\mathrm{QFT}_n(|k\rangle) = \frac{1}{\sqrt{n}}\sum_{j=0}^{n-1} e^{-2\pi ijk/n}|j\rangle$$

Exercise

A state space $|\mathbb{Z}_n\rangle$ generated by n orthogonal computational states $|0\rangle, |1\rangle, \dots |n-1\rangle$ should not seem like a departure from the binary states $|0\rangle, |1\rangle, \frac{|00\rangle}{\sqrt{2}} + \frac{|11\rangle}{\sqrt{2}}$, etc. that are most common in programming. Show that any superposition $|\psi\rangle = c_0|0\rangle + \dots c_{n-1}|n-1\rangle$ can be represented using $N = \lceil \log_2(n) \rceil$ registers of ordinary $|0\rangle, |1\rangle$ qubits. Explain why any unitary operation U on $|\mathbb{Z}_n\rangle^{\otimes m}$, which is to say any unitary operation on m registers each drawn from the space $|\mathbb{Z}_n\rangle$, can be embedded within a unitary U acting on mN registers of ordinary $|0\rangle, |1\rangle$ qubits.

Given the *periodic* nature of sinusoids and this kind of frequency analysis, it should feel perfectly intuitive that the DFT / QFT is none other than the Group Fourier Transform corresponding to the *cyclic* group $\mathbb{Z}_n$[3]. After all, the GQFT should at least *generalize* the FT, and we've already noticed the exponential $e^{-2\pi ik/n}$ appearing as a representation of $\mathbb{Z}_n$.

Proof: We opened with the example of $\rho_1(k) = [e^{-2\pi ik/n}]$ being a representation of $\mathbb{Z}_n$, and the same is clearly true for any $\rho_j(k) = [e^{-2\pi ijk/n}]$ with $0 \le j \le n-1$, giving us a total of n distinct representations. It's trivial to see that these are all unitary (take the complex conjugate), irreducible ($d = 1$ so clearly no redundant sub-spaces can exist), and inequivalent to each other (a similarity transform just cancels), so in fact we've produced at least n elements of $\widehat{\mathbb{Z}_n}$. But of course, $n = |\mathbb{Z}_n| = \sum_\rho d_\rho^2 \ge \sum_j 1^2 = n$ so we've actually found all of them.

The utility of the Discrete/Quantum Fourier Transform centers around a classic *filtration* identity: $\frac{1}{n}\sum_{j=0}^{n-1} e^{2\pi ija/n}e^{-2\pi ijb/n} = \delta_{a=b}$, which elegantly weeds out the appearance of cross terms using deconstructive interference of the complex exponentials. Now, recall the column orthogonality formula for characters, ie 1, and recall that since $\mathbb{Z}_n$ is Abelian, its conjugacy classes are just singletons surrounding each element of the group. Is it not beautiful to observe that the above filtration identity $\sum_{\rho_j\in\widehat{G}} \overline{e^{-2\pi iaj/n}}e^{-2\pi ibj/n} = \sum_{j=0}^{n-1} e^{2\pi iaj/n}e^{-2\pi ibj/n} = \begin{cases} 0 & a \ne b \\ \frac{n}{1} & a = b \end{cases}$ falls naturally out of studying these group characters?

Exericse

Suppose ρ is a 1×1 representation of $\mathbb{Z}_n$. What should $\rho(1)$ be given that $\rho(1)^n = \rho(n) = \rho(0)$? In this way, one does not have to "guess" what the irreducible representations are. What happens when $n = 2$? Establish that the humble Hadamard gate H is in fact the QGFT of $\mathbb{Z}_2$: the foundational group for working with binary arithmetic. Carefully compute some elements of the matrix QFT_4: is it equal to $H \otimes H$? Why or why not?

Having established $\mathcal{F}_{\mathbb{Z}_n} = \mathrm{QFT}_n$, the most intuitive direction towards expanding our toolkit would be to identify the QGFT for groups composed of this fundamental cyclic building-block: for example $\mathbb{Z}_n \times \mathbb{Z}_m$ or the space of binary vectors $\mathbb{Z}_2^n$, or more generally any finite group of the form $Z = \mathbb{Z}_{n_1}^{m_1} \times \mathbb{Z}_{n_k}^{m_k}$. It is natural to think that given a quantum register housing the system $|\mathbb{Z}_{n_1}\rangle$, a register housing $|\mathbb{Z}_{n_2}\rangle$, and so on up to $|\mathbb{Z}_{n_k}\rangle$, that the QGFT associated with the space of group elements $|\mathbb{Z}_{n_1} \times \ldots \mathbb{Z}_{n_k}\rangle$ should correspond to applying a QFT_{n_1} gate to the first register, a QFT_{n_2} gate to the second register and so on as it is needed.

This construction is in fact correct![2] For any group $Z = \mathbb{Z}_{n_1}^{m_1} \times \ldots \mathbb{Z}_{n_k}^{m_k}$ we have $\mathcal{F}_Z = \mathrm{QFT}_{n_1}^{\otimes m_1} \otimes \ldots \mathrm{QFT}_{n_k}^{\otimes m_k}$, which corresponds to applying m_1 $n_1 \times n_1$ QFT gates, m_2 $n_2 \times n_2$ QFT gates, etc. all in *parallel* to a quantum system consisting of $|Z| = n_1^{m_1} \ldots n_k^{m_k}$ computational basis states. The proof that the direct product $\times$ of these cyclic groups translates to the tensor product $\otimes$ of the corresponding QGFT gates is yet another beautiful result at the intersection of quantum computing and representation theory:

Multiplicativity of the QGFT

Let A and B be finite groups. Every $\rho \in \widehat{A \times B}$ can be uniquely expressed as $\rho(a, b) = \rho_A(a) \otimes \rho_B(b)$ for some $\rho_A \in \widehat{A}, \rho_B \in \widehat{B}$, and conversely the tensor product of all such distinct pairs ρ_A, ρ_B generates all of the unitary, irreducible, inequivalent representations of $A \times B$.

Using the natural coordinates produced by this identification, the QGFT is *multiplicative*:

$$\mathcal{F}_{A\times B} = \mathcal{F}_A \otimes \mathcal{F}_B \tag{7}$$

Proof: By "natural coordinates" we simply mean that $\left|\widehat{A \times B}\right\rangle$ will be generated by the computational basis states $|\rho_{A,i,j}, \rho_{B,k,\ell}\rangle$ as $1 \le i, j \le d_{\rho_A}$ and $1 \le k, \ell \le d_{\rho_B}$ range over their usual parent states, as this allows us to very easily identify $|\rho_{A,i,j}, \rho_{B,k,\ell}\rangle$ with $|\rho_{A,i,j}\rangle \otimes |\rho_{B,k,\ell}\rangle$ and hence make good progress when we expand the tensor product:

$$(\mathcal{F}_A \otimes \mathcal{F}_B) |a\rangle \otimes |b\rangle = \left(\sum_{\rho_A \in \widehat{A}} \sqrt{\frac{d_{\rho_A}}{|A|}} \sum_{1 \le i,j \le d_{\rho_A}} [\rho_A(a)]_{i,j} |\rho_{A,i,j}\rangle \right) \otimes \left(\sum_{\rho_B \in \widehat{B}} \sqrt{\frac{d_{\rho_B}}{|B|}} \sum_{1 \le k,\ell \le d_{\rho_B}} [\rho_B(b)]_{k,\ell} |\rho_{B,k,\ell}\rangle \right) =$$

$$\sum_{(\rho_A,\rho_B) \in \widehat{A} \times \widehat{B}} \sqrt{\frac{d_{\rho_A} d_{\rho_B}}{|A||B|}} \sum_{\substack{1 \le i,j \le d_{\rho_A} \\ 1 \le k,\ell \le d_{\rho_B}}} [\rho_A(a) \otimes \rho_B(b)]_{i,j;k,\ell} |\rho_{A,i,j}, \rho_{B,k,\ell}\rangle$$

Calling $\rho = \rho_A \otimes \rho_B$ it's clear that the inner-most sum runs over all of the entries in ρ, as is the fact that $d_\rho = d_{\rho_A} \times d_{\rho_B}$ and of course $|A \times B| = |A||B|$, so together with the representation-theoretic assumption that $\widehat{A} \times \widehat{B} \cong \widehat{A \times B}$, it's clear we arrive at:

$$\sum_{\rho \in \widehat{A \times B}} \sqrt{\frac{d_\rho}{|A \times B|}} \sum_{1 \le i',j' \le d_\rho} [\rho(a, b)]_{i',j'} |\rho_{i',j'}\rangle = \mathcal{F}_{A \times B}(a, b)$$

Where again, we must ensure that we sum over the entries $\rho_{i',j'}$ of ρ is taken in accordance with the tensor product of ρ_A and ρ_B. Now, based of a quick check like $[\rho_A(a) \otimes \rho_B(b)][\rho_A(a') \otimes \rho_B(b')] =$

$\rho_A(a)\rho_A(a') \otimes \rho_B(b)\rho_B(b') = \rho_A(aa') \otimes \rho_B(bb')$, it's not hard to see that combining $\rho_A \otimes \rho_B$ with a tensor product gives a valid representation, and to show that it's an irreducible representation, we make use of the $\langle \chi_\rho, \chi_\rho \rangle = 1$ criterion:

$$\langle \chi_\rho, \chi_\rho \rangle = \frac{1}{|A \times B|} \sum_{g \in A \times B} \overline{\chi_\rho(g)} \chi_\rho(g) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} |\mathrm{tr}(\rho_A(a) \otimes \rho_B(b))|^2$$

$$= \left(\frac{1}{|A|} \sum_{a \in A} |\mathrm{tr}(\rho_A(a))|^2 \right) \left(\frac{1}{|B|} \sum_{a \in B} |\mathrm{tr}(\rho_B(a))|^2 \right) = \langle \chi_{\rho_A}, \chi_{\rho_A} \rangle \langle \chi_{\rho_B}, \chi_{\rho_B} \rangle = (1)(1) = 1$$

Where we have used the fact that ρ_A and ρ_B are themselves irreducible, and employed another classic identity of the trace: $\mathrm{tr}(P \otimes Q) = \mathrm{tr}(P)\mathrm{tr}(Q)$ which is to say that $\chi_{\rho_A \otimes \rho_B} = \chi_{\rho_A} \chi_{\rho_B}$ conveniently factorizes. We'll use this factorization an additional time to establish that if ρ_A and ρ'_A are two inequivalent representations, and same for ρ_B and ρ'_B, then their products $\rho_A \otimes \rho_B$ and $\rho'_A \otimes \rho'_B$ remain inequivalent, with the relevant criterion from χ being:

$$\langle \chi_{\rho_A \otimes \rho_B}, \chi_{\rho'_A \otimes \rho'_B} \rangle \neq 0 \Leftrightarrow \frac{1}{|A \times B|} \sum_{(a,b) \in A \times B} \overline{\chi_{\rho_A}(a) \chi_{\rho_B}(b)} \chi_{\rho'_A}(a) \chi_{\rho'_B}(b)$$

$$= \left(\frac{1}{|A|} \sum_{a \in A} \overline{\chi_{\rho_A}(a)} \chi_{\rho'_A}(a) \right) \left(\frac{1}{|B|} \sum_{b \in B} \overline{\chi_{\rho_B}(b)} \chi_{\rho'_B}(b) \right) = \underbrace{\langle \chi_{\rho_A}, \chi_{\rho'_A} \rangle}_{\neq 0} \underbrace{\langle \chi_{\rho_B}, \rho'_B \rangle}_{\neq 0}$$

So, we've concluded that each choice of $\rho_A \in \widehat{A}$ and $\rho_B \in \widehat{B}$ provides us with an irreducible, inequivalent element $\rho_A \otimes \rho_B$ of $\widehat{A \times B}$ and like before we can show our list is an exhaustive enumeration of $\widehat{A \times B}$ by checking whether the sum $\sum_\rho d_\rho^2$ matches the size of $|A \times B|$. This involves one more factorization:

$$|A \times B| = \sum_{\rho \in \widehat{A \times B}} d_\rho^2 \geq \sum_{\rho_A \in \widehat{A}, \rho_B \in \widehat{B}} \underbrace{d^2_{\rho_A \otimes \rho_B}}_{d^2_{\rho_A} d^2_{\rho_B}} = \left(\sum_{\rho_A \in \widehat{A}} d^2_{\rho_A} \right) \left(\sum_{\rho_B \in \widehat{B}} d^2_{\rho_B} \right) = |A||B|$$

The $\times \leftrightarrow \otimes$ correspondence is particularly intuitive: the group-theoretic notion of $A \times B$ as the *independent* combination of both group structures where the action $(a, b) \circ (a', b') = (a \circ a', b \circ b')$ is taken *in parallel* corresponds perfectly to *independently* applying the QGFT for A to some portion of the quantum registers and applying the QGFT for B to the remaining registers in *parallel*.

Notice that in particular, a huge quantum advantage is had[3] when $n \ll m$ given that applying m small, identical QFT gates all in parallel is trivially simple compared to implementing a high-depth-complexity QFT gate on a large register for large n: this makes for an obvious but important practical distinction between the $\mathbb{Z}_{2^n}$ (single gate) and $\mathbb{Z}_2^n$ (parallelized) groups despite both being of equal size.

In fact, every time in class that we've applied a layer of Hadamard gates to a set of registers in parallel, we've been making good of use of this effect[4]: we've been computing $2^n \times 2^n$ QGFT's without blinking an eye!

Hadamard Transform

The Hadamard Transform $\mathcal{H}$ on a register of n qubits is simply $H^{\otimes n}$, ie applying the H gate in parallel to n registers. Mathematically, given a binary string $b = b_1, \ldots b_n$, this takes the form:

$$\mathcal{H}(|x\rangle) = \frac{1}{\sqrt{2^n}} \sum_{b_1, \ldots b_n \in \{0,1\}} (-1)^{b_1 x_1 + \ldots b_n x_n} |b\rangle$$

Those n qubit registers characterize the group $\mathbb{Z}_2^n$ and so logically:

$$\mathcal{H} = \mathcal{F}_{\mathbb{Z}_2^n}$$

Proof: Given that $\mathbb{Z}_2^n$ is Abelian, the representation-theoretic machinery needed for the general theorem above is laughably overcomplicated compared to this special case: $\mathbb{Z}_2$ consists of the identity and a single non-trivial element which squares to the identity, while $\widehat{\mathbb{Z}_2}$ consists of the identity map $|x_i\rangle \mapsto |x_i\rangle$ and the negation map $|x_i\rangle \mapsto -|x_i\rangle$ which squares to the identity map. Hence, to each $|b_i\rangle$ in the ith copy of $|\mathbb{Z}_2\rangle$ appearing in the tensor product $|\mathbb{Z}_2^n\rangle \cong |\mathbb{Z}_2\rangle \otimes \dots |\mathbb{Z}_2\rangle$ we associate the linear map $|x_i\rangle \mapsto (-1)^{b_i x_i} |x_i\rangle$, which tensors together into $|b_1, \dots b_n\rangle$ being associated the linear map $(-1)^{b_1 x_1} |x_1\rangle \otimes \dots (-1)^{b_n x_n} |x_n\rangle = (-1)^{b_1 x_1 + \dots b_n x_n} |x_1 \dots x_n\rangle$. Since every representation here is of dimension 1, it's not exactly hard to see why each element $|\rho\rangle$ of $\left|\widehat{\mathbb{Z}_2^n}\right\rangle$ is conveniently associated with an element $|b\rangle$ of $|\mathbb{Z}_2^n\rangle$, and so with a slight abuse of notation, the identification between $\mathcal{H} : \mathbb{Z}_2^n \to \mathbb{Z}_2^n$ and $\mathcal{F}_{\mathbb{Z}_2^n} : |\mathbb{Z}_2^n\rangle \to \left|\widehat{\mathbb{Z}_2^n}\right\rangle$ is obvious. It should also *feel* obvious from the circuit point of view since recall the QGFT for $\mathbb{Z}_2$ was the ordinary Hadamard gate H and the transform $\mathcal{H}$ is the simultaneous application of n of those H gates.

Recall also that in algorithms such as Bernstein-Vazirani and Simon's Problem, the key mechanism which reduced an exponentially large superposition to a "concentrated" one involving only the desirable states was the following *filtration identity* for the Hadamard transform:

$$\frac{1}{2^n}\sum_{b\in\mathbb{Z}_2^n}(-1)^{b\cdot x}(-1)^{b\cdot y} = \delta_{x=y} \quad \text{vs.} \quad \frac{1}{N}\sum_{n\in\mathbb{Z}_N} e^{2\pi i n/Nx} e^{-2\pi i n/Ny} = \delta_{x=y} \quad \text{vs.} \quad \frac{1}{|G|}\sum_{\rho\in\widehat{G}}\overline{\chi_\rho(g)}\chi_\rho(h) = \delta_{g=h}$$

On the left. Take a moment to appreciate that the two most important filtration identities, the workhorse of quantum interference, are in fact special cases of the more general *character orthogonality* relation seen on the right, for any Abelian group G. Equivalently, it's not hard to see that these "filtration identities" express exactly the same information as $\mathcal{H}, \text{QFT}, \mathcal{F}_G$, etc, being *unitary operators*, and this might be a more intuitive notion given the quantum context.

Even more excitingly, it is a classic and well-known fact of Group Theory (the Fundamental Theorem of Finitely Generated Abelian Groups[1]) that *every* finite *Abelian* group G is isomorphic to $\mathbb{Z}_{n_1}^{m_1} \times \dots \mathbb{Z}_{n_k}^{m_k}$, with n_i, m_i determined uniquely up to shuffling around the factors and possibly combining terms in a way that is not needed to study here.

The intuitive significance of this decomposition is then that for *any* abelian group G we can both compute and reasonably quickly implement[3] the QGFT $\mathcal{F}_G$, giving us a "bespoke" filtration identity tailored to support quantum interference in "the right way". A more formal approach is given in the next section.

Exercise

Suppose that $\rho_A \otimes \rho_B$ and $\rho'_A \otimes \rho'_B$ are inequivalent: using standard properties of the $\otimes$ product, show that ρ_A and ρ'_A are inequivalent as are ρ_B and ρ'_B. Unfortunately going the other direction is not so trivial, hence why a more indirect χ-based proof was used.

Statement of the Hidden Subgroup Problem

It is impossible to discuss the importance of the QGFT without formalizing what problem it is "meant" to solve [5]. Although a bit un-motivated, we ended off 5 by quantifying a nice ability of the QGFT to extract information about a subgroup H given only access to a *coset* state $|gH\rangle$, and it is this capacity which makes it inseparable from the following class of problems:

Hidden Subgroup Problem (HSP)

Let G be a finite group and let $H \leq G$. Given to you is a function $f : G \to \Sigma$ (the range can be anything and is not terribly important) which labels whether two elements of G are representatives of the same left coset of H. Formally, for any $g, g' \in G$ we have $f(g) = f(g')$ *if and only if* $gH = g'H$, or equivalently $g'^{-1}g \in H$. Using as few calls to the oracle f as possible, recover a set of generators for H.

Such a *coset identification* function f might sound completely unnatural to come across in practice, but in reality it is sufficient to find *any* group *homomorphism* $f : G \to \Sigma$ with $\ker(f) = H$, and this is simply because $gH = g'H \Leftrightarrow g'^{-1}g \in H \Leftrightarrow f(g'^{-1}g) = e \Leftrightarrow f(g')^{-1}f(g) = e \Leftrightarrow f(g') = f(g)$; which is of course the First Isomorphism Theorem. In fact, all of the examples of f in this note take this form, but in general the HSP does *not* demand any special properties of f (like being a homomorphism) outside of respecting the cosets of G/H.

Remarkably, this rather vague notion of piecing together subgroups from cosets underlies nearly all of the most famous and important quantum algorithms: Simon's Problem, Shor's Factoring Algorithm, the Discrete Logarithm Problem, Bernstein-Vazirani, Hidden Slope, Order Finding, and many, many more beyond the scope of PHYS 3450. What's particularly beautiful about the above formalism is that once the function f has been found or constructed, there is a nearly *uniform* procedure for executing the quantum algorithm to extract *constraints* on the subgroup H: apply $\mathcal{F}_G$ to an empty register, query the function using a unitary oracle U_f on some ancilla qubits, measure the resulting ancillas, then apply $\mathcal{F}_G^\dagger$ to close the quantum interference, and finally record the values of the primary register.

As you established in the last exercise of the first section, applying $\mathcal{F}_G$ to an empty register of $|0\dots0\rangle$ generates the uniform superposition over group states . Then, upon querying and later observing the value of f acting on an empty ancilla register, we reduce this uniform superposition down to one of size $|H|$:

$$\frac{1}{\sqrt{|G|}}\sum_{g\in G}|g\rangle \overset{U_f}{\rightsquigarrow} \frac{1}{\sqrt{|G|}}\sum_{g\in G}|g\rangle\otimes|0\oplus f(g)\rangle \overset{\text{Measure } f(g_0)}{\rightsquigarrow} \frac{1}{\sqrt{|H|}}\sum_{g\in G: f(g)=f(g_0)}|g\rangle\otimes|f(g_0)\rangle = \frac{1}{\sqrt{|H|}}\sum_{h\in H}|g_0h\rangle \tag{8}$$

Take note of how the promise on f means that *measuring* a certain value from its image naturally collapses the state down to a particular *coset* state in the pre-image whose representative g_0 say is unknown. The QGFT can then be used to remove the influence of g_0 as we confirmed back in 6 and will continue to explore in the next section.

Here's another elementary definition: the action of a group G on some "object" x in a "family" of similar objects X is simply a map $x \mapsto g \circ x \in X$ which is *compatible* with the group action of G: the identity element of G must fix each $x \in X$ and composing two group actions together $g_1g_2 \circ x$ should correspond to, clearly, composing the action of $g_2 \circ x$ followed by the action of g_1, ie $g_1 \circ (g_2 \circ x)$. For example, $\mathbb{Z}_2$ might act on the $\mathbb{R}^2$ plane by reflection about the horizontal axis (two reflections give you back to the identity), the symmetric group S_3 might act on the vertices of a triangle by permuting them, or the circle group S^1 might act on the unit disk $|z| \leq 1 \subseteq \mathbb{C}$ by rotating it: in each case the action of the group on the object simply corresponds to a set of bijections of X whose composition reflects the composition of elements in G.

Interestingly, group actions give rise to *incredibly natural* examples of HSPs:[5] the coset indicator function in question is simply "the action", namely $f : G \to X$ given by $f(g) = g \circ x$ where the hidden subgroup in question is $H = \mathrm{stab}_G(x)$: the subgroup of all elements of G which leave x

unchanged ie $g \circ x = x$. Indeed, the Orbit Stabilizer Theorem tells us that the cosets of H in G correspond precisely with the unique possible images that x can get sent to under G (the "orbit" of x), but more concretely it's not hard to verify the coset-identification property algebraically with $g \circ x = g' \circ x \Leftrightarrow g'^{-1} g \circ x = x \Leftrightarrow g'^{-1} g \in \mathrm{stab}_G(x) = H \Leftrightarrow g'H = gH$.

Since every group action corresponds to a particular permutation representation of some subgroup of S_n, this makes the *Automorphism Problem*, namely the task of enumerating which permutations of S_n leave a particular object fixed, another important and natural instance of the HSP framework. In particular, the *Graph Automorphism* problem, determining which permutations of the vertices of a graph leave its edges symmetrically unchanged, is an active area of HSP research due to its possibly exponential advantage over classical techniques.

This note is *not* meant to explore the HSP in much detail (or focus remains on the QGFT), but there are two fundamentally important notions necessary to proceed[3]:

- **Abelian vs. Non-Abelian HSP:** It turns out, although we do not prove it here, that the ability to construct the QGFT for any arbitrary Abelian group (recall that this was of the form $\mathrm{QFT}_{n_1}^{\otimes m_1} \otimes \ldots \mathrm{QFT}_{n_k}^{\otimes m_k}$) permits an exact reconstruction of the hidden subgroup H in poly-logarithmic (in terms of $|G|$) number of calls to the above sampling routine. We sketch some intuition in the proceeding exercise, but for more information see [2].

  However, as soon as the promise for G being Abelian is lost, the process of reconstruction can become much more complex, as we try to sketch loosely in the last section. In particular, their are known [5] instances of the Automorphism Problem (recall, a subgroup of S_n and hence likely non-Abelian) which, following our *naive* measurement process above, produce exponentially similar distributions of recorded representations ρ: indicating, at least philosophically, a fundamental difficulty with *distinguishing* subgroups of non-Abelian groups G via the QGFT method.

- **Weak vs. Strong Fourier Sampling:** When G is non-Abelian there is an important information-theoretic boundary between what can be reconstructed from our formula 6 regarding the probability of measuring *some* Fourier-representation mode of ρ and the probability distribution of measuring an *individual* mode $|\rho_{i,j}\rangle$. The former case, aggregated over the $|\rho_{i,j}\rangle$'s, is known as *Weak Sampling* and is the simpler notion this note is based around, with the latter case being *Strong Sampling*, where information about which entries in ρ may be used to help in the H-recovery process.

  Depending on the particular oracle implementation U_f and the QGFT, it's possible that upon measuring the ancilla registers one simply does not have access to a specific $|\rho_{i,j}\rangle$ and only the aggregated probabilities for $P(\rho)$, and as we will exemplify in 1, there are instances of the HSP that are considerably more difficult when keeping track of only the *weakly-sampled* ρ values. As per usual, we defer the details to [2].

Exercise

It is a classic fact of representation theory that if G is Abelian, its irreducible representations always have dimension $d_\rho = 1$. Considering our definition 2, what simplifications occur? In particular, what is χ_ρ? Going all the way back to 1, what is the size of the conjugacy class of a $g \in G$? What does "strong" vs "weak" Fourier sampling look like in this case? For Abelian G, one writes $H^\perp$ for the set of $\rho \in \widehat{G}$ such that $\rho(h) = 1$ for all $h \in H$. With this in mind, compute:

$$\frac{1}{|H|} \sum_{h \in H} \chi_\rho(h)$$

Intuitively speaking, when we measure from 8, from what set of representations are we primarily sampling? If $|H|$ is large relative to $|G|$, what will the distribution of measured ρ's look like?

Two Famous (Abelian) Applications

In this section our goal is to put the tools and theory developed so far regarding the HSP-QGFT pairing to the test. We will walk through the QGFT-based algorithmic solutions to 2 of the most famous[4] Abelian HSP's, namely the "Hidden Slope" or "Hidden Linear Functional Problem" (a family of problems making repeated in-class appearances) and the Discrete Logarithm Problem (of foundational importance to cryptographic security). For each, take a moment to reflect on *how* each problem encodes a cosets of a hidden subgroup and *why* the gates specified should actually be applied, drawing from the theory of QGFT. Let's dive in!

Hidden Slope Problem

Let $f : \mathbb{Z}_n^k \to \mathbb{Z}_n$ be a promised affine function $f(x) = a^\top x + b$ where $a \in \mathbb{Z}_n^k, b \in \mathbb{Z}_n$ and all multiplications and addition are taken modulo n. A standard for oracle for f is provided in the form $U_f \left|x_1, \ldots x_k\right\rangle \left|y\right\rangle = \left|x_1, \ldots x_k\right\rangle \left|y + f(x_1, \ldots x_k) \pmod n\right\rangle$ where $\left|x_i\right\rangle$ and $\left|y\right\rangle$ are among the Hilbert space generated by the computational states $\left|0\right\rangle, \ldots \left|n-1\right\rangle$. Using as few calls to U_f as possible, determine the k elements of $\mathbb{Z}_n$ that make up a (the "slope" of f).

Before anything else, it's important to build familiarity with what function f is being promised to us. For example, when $n = 2$ and $k = 1$, there are 4 possible affine functions from $\mathbb{Z}_2 \to \mathbb{Z}_2$ namely $0x + 0 = 0, 0x + 1 = 1, 1x + 0 = x, 1x + 1 = \neg x$ which leads us directly into *Deutsch's Algorithm*! Determining whether the slope a is 0 or 1 is tantamount to determining whether f is constant or balanced, and so the "Slope Problem" generalizes the former. The same reduction is clearly true of the *Bernstein–Vazirani* algorithm too: when $n = 2$ and $k \geq 1$, finding the secret bit string s such that $f(x_1, \ldots x_k) = s_1 x_1 + \ldots s_k x_k \pmod 2$ is clearly identical to finding the "slope" of the affine function $s^\top x + 0$ over $\mathbb{Z}_2$. Recall that the Deutsch Algorithm opened and closed quantum interference using a Hadamard gate H and that Bernstein–Vazirani did the same with a layer of k simultaneous Hadamards, ie $H^{\otimes k}$: it should come as no surprise that these transformations are the QGFT respectively of $\mathbb{Z}_2^1$ and $\mathbb{Z}_2^k$.

Recall that in both of these special cases, the oracle U_f was applied not to an empty ancilla register containing $\left|0\right\rangle$ but rather containing $\left|-\right\rangle$: which has the handy property of making the input $\left|x_1, \ldots x_k\right\rangle \left|-\right\rangle$ an *eigenstate* of U_f since of course the result would be $\left|\mathbf{x}\right\rangle \frac{\left|f(\mathbf{x}) \oplus 0\right\rangle - \left|f(\mathbf{x}) \oplus 1\right\rangle}{\sqrt{2}} = \left|\mathbf{x}\right\rangle (-1)^{f(\mathbf{x})} \left|-\right\rangle$. In this form, the data from f is encoded in *phase*, which is essential for the process of deconstructive interference to occur[5].

More generally, when the elements of our quantum system are not simply bits $\left|0\right\rangle, \left|1\right\rangle$ but rather $\left|0\right\rangle, \ldots \left|n-1\right\rangle$ pure states of $\left|\mathbb{Z}_n\right\rangle$, we have an analogous helper state defined by $\left|\sim\right\rangle = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} e^{2\pi i k/n} \left|k\right\rangle = \mathcal{F}_{\mathbb{Z}_n}(\left|1\right\rangle)$ whose purpose is to ensure that $U_f \left|\mathbf{x}\right\rangle \left|\sim\right\rangle = \left|\mathbf{x}\right\rangle \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} e^{2\pi i k/n} \left|k + f(\mathbf{x}) \pmod n\right\rangle = \left|\mathbf{x}\right\rangle e^{-2\pi i f(\mathbf{x})/n} \left|\sim\right\rangle$ acts according to a transformation of phase. For example, when $n = 2$ we recover the familiar $\frac{1}{\sqrt{2}}(e^{\pi i 0} \left|0\right\rangle + e^{\pi i 1} \left|1\right\rangle) = \left|-\right\rangle$.

It is extremely easy to understand the quantum algorithm for the Hidden Slope Problem without any appeal to HSP theory, however for completeness, it's worth identifying what subgroup of $\mathbb{Z}_n^k$ we're actually looking for: does $f : \mathbb{Z}_n^k \to \mathbb{Z}_n$ provide us with a homomorphism? After removing the irrelevant bias, the answer is certainly yes: clearly for any two $x, x' \in \mathbb{Z}_n^k$ we have $a^\top(x + x') = a^\top x + a^\top x'$ and so f provides a coset identification function for the subgroup $H = \ker a = \{x \in \mathbb{Z}_n^k : a^\top x = 0\}$, or rather the null-space of the linear transformation $x \mapsto a^\top x$. Since f is a surjection onto $\mathbb{Z}_n$, it's not hard to check that $\operatorname{img} f \cong G/H \implies n = |\mathbb{Z}_n^k|/|H| \implies |H| = n^{k-1}$. According to our mantra of "one measurement of $\mathcal{F}_G$ gives one constraint on H", it's at least *plausible* that we should be able to recover H in exactly one query since H itself is defined by carving out the subspace of $\mathbb{Z}_n^k$ satisfied by a *single* linear constraint.

The algorithm proceeds in the canonical "$\mathcal{F}_G U_f \mathcal{F}_G^\dagger$" fashion described in the previous section: apply $\mathcal{F}_{\mathbb{Z}_n^k}$ to a primary register containing $\left|0\right\rangle^{\otimes k}$ and then query the oracle U_f by acting on the state $\left|\sim\right\rangle$ in an ancilla register. After observing/measuring the output state of the ancilla register, apply $\mathcal{F}_{\mathbb{Z}_n^k}^\dagger$

to the primary register and read out the k elements of the slope a. As we know, applying $\mathcal{F}_G$ to an empty register of $|0\dots0\rangle$'s simply prepares a uniform superposition over each of the elements of our group:

$$\frac{1}{\sqrt{|\mathbb{Z}_n^k|}}\sum_{j_1=0}^{n-1}\cdots\sum_{j_k=0}^{n-1}|j_1,\dots j_k\rangle\otimes|\sim\rangle$$

Then, upon applying the oracle we understand that the action of f on $|\sim\rangle$ takes the form of a phase factor:

$$\frac{1}{\sqrt{|\mathbb{Z}_n^k|}}\sum_{j_1=0}^{n-1}\cdots\sum_{j_k=0}^{n-1}e^{-2\pi i/nf(j_1,\dots j_k)}|j_1,\dots j_k\rangle\otimes|\sim\rangle=\frac{1}{\sqrt{|\mathbb{Z}_n^k|}}\sum_{j_1=0}^{n-1}\cdots\sum_{j_k=0}^{n-1}e^{-2\pi i/na_1j_1}\cdots\times e^{-2\pi i/na_kj_k}\times e^{-2\pi i/nb}|j_1,\dots j_k\rangle\otimes|\sim\rangle$$

Which, after removing the irrelevant $e^{-2\pi i/nb}$ global phase factor at the end, clearly takes the form of a k-fold tensor product of *independent* registers each in the state $\frac{1}{\sqrt{|\mathbb{Z}_n|}}\sum_{j=0}^{n-1}e^{-2\pi i/naj}|j\rangle$. Measuring $|\sim\rangle$ clearly doesn't disrupt anything.

Of course, since $\mathcal{F}_{\mathbb{Z}_n}(|a\rangle)=\frac{1}{\sqrt{|\mathbb{Z}_n|}}\sum_{j=0}^{n-1}e^{-2\pi i/naj}|j\rangle$, it's clear that upon applying $\mathcal{F}_{\mathbb{Z}_n}^\dagger$ to each of the k registers we will directly return to the pure $|a\rangle$ state awaiting our measurement. Indeed, since each register behaves independentantly and identically in this way, and since $\mathcal{F}_{\mathbb{Z}_n^k}^\dagger=\mathcal{F}_{\mathbb{Z}_n}^\dagger\otimes\dots\mathcal{F}_{\mathbb{Z}_n}^\dagger$, the result of closing the interference with the QGFT is a set of registers in the pure $|a_1,\dots a_k\rangle$ state: our secret slope!

Exercise

Eve promises Alice a function $f:\mathbb{Z}_n^k\to\mathbb{Z}_n^\ell$ given by $f(x)=Ax+b$ where A is an $\ell\times k$ matrix and b is an ℓ-vector, again always taken $\pmod n$. Describe an approach to determine all ℓk elements of $\mathbb{Z}_n$ which make up the matrix A using ℓ calls to the oracle U_f (which now requires ℓ output registers) as well as how the vector b can be obtained. Assuming Eve measures the ancillas and Alice can only observe the k primary registers, show that Alice can gain at most $k\log_2(n)$ *classical* bits of information from each measurement of f. How many bits of information are required to specify ℓk elements of $\mathbb{Z}_n$? Considering the Holevo Bound, is it possible to extract all of A in fewer than ℓ calls to f?

Now, Shor's Algorithm for the discrete logarithm problem, like with the closely related Shor's Factoring Algorithm and the more distantly related Simon's Problem, but *unlike* with the hidden slope problem above, will involve measurement outcomes which might not perfectly specify the desired hidden subgroup at hand[2]: but this is no issue given that simply re-running the algorithm provides an exponentially low probability of repeated "failure". Despite it's profound importance to cryptography, the Discrete Logarithm Problem (DLP) in a group A is remarkably easy to state:

Discrete Logarithm Problem

Let $A=\langle\alpha\rangle$ be a cyclic group of order N with a known generator α (meaning every element of A is some power of α with α^N looping back to 1). The *Discrete Logarithm* of some $h\in A$ is the unique value of $0\le n<N$ such that $\alpha^n=h$.

In this way one could write "$n=\log_\alpha(h)$" but do not be deceived: calculation of this quantity is not at all like the explicit, real-valued logarithm given that the periodicity of the generator α typically forms a "random" walk through the elements of A. Similarly, do not be deceived by the appearance of the cyclic group $\mathbb{Z}_N\cong A$: although we will be making use of QFT_N, Shor's DLP algorithm actually cleverly takes place within the group $\mathbb{Z}_N\times\mathbb{Z}_N$.

This is because we are given *two* pieces of group data from A, each having order dividing N: namely the known generator α and the target value $h=\alpha^n$, which allows us to furnish a well-defined homomorphism $f:\mathbb{Z}_N\times\mathbb{Z}_N\to G$ given by $(a,b)\mapsto\alpha^ah^b$. And, indeed, the commutativity of A tells us that $(a,b)+(a',b')\mapsto\alpha^ah^b\alpha^{a'}h^{b'}=\alpha^{a+a'}h^{b+b'}=f(a+a',b+b')$: doing this calculation also shows us that for any integer k, the specific tuple $(nk, -k) \mapsto \alpha^{nk}h^{-k} = \alpha^{nk}\alpha^{-nk} = 1$ lies within the kernel of the homomorphism f and indeed the hidden subgroup H will be given by $\ker f = \langle (n, -1) \rangle$, ie the cyclic subgroup generated by the tuple whose first entry is precisely the value of n we're looking for!

Although how you might come up with such a map f in the first place is unimportant to the algorithm, I would like to make two remarks about the characteristics of f which could provide some motivation:

- Abelian groups G are often studied abstractly as a $\mathbb{Z}$-module: essentially a "vectorspace", or more cryptographically speaking a *lattice*, where addition and scalar multiplication by elements of $\mathbb{Z}$ are isomorphically encoded by multiplying generators of G. If every element of G can be written as a commutative product of $g_1^{a_1} \dots g_k^{a_k}$ for some generators $g_1, \dots g_k \in G$, then just like we saw above, addition of "vectors" $(a_1, a_2, \dots a_k) + (a'_1, a'_2, \dots a'_k)$ corresponds to composition of group elements $g_1^{a_1} \dots g_k^{a_k} \circ g_1^{a'_1} \dots g_k^{a'_k} = g_1^{a_1+a'_1} \dots g_k^{a_k+a'_k}$ and the same is true for powers $(g_1^{a_1} \dots g_k^{a_k})^b \Leftrightarrow b(a_1, \dots a_k)$ too, which you can verify satisfies the distributive law, etc.

  The above map f is therefore just the canonical method of converting between an Abelian group and its $\mathbb{Z}$-module counterpart. Also, hopefully this $\mathbb{Z}$-module picture gives some credence to the notion that all Abelian groups really do have a $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \dots \mathbb{Z}_{n_k}$ structure[1]: the resulting $\mathbb{Z}$-module has something of a "basis" where each group element g_i has order n_i, generating its own copy of $\mathbb{Z}_{n_i}$.

- The famous *Birthday Attack* on the Discrete Logarithm involves an attacker Eve forming two large pools of elements of A, namely $\{\alpha^{n_1}, \alpha^{n_2}, \dots\}$ containing uniformly sampled powers of α and $\{h\alpha^{m_1}, h\alpha^{m_2}, \dots\}$ again with sampled powers but now scaled by the public parameter h. As we know, it is expected to take around $O(\sqrt{N})$ elements in these lists in order for a collision, namely $\alpha^{n_i} = h\alpha^{m_j} \implies h = \alpha^{n_i - m_j} \implies \log_\alpha(h) = n_i - m_j \pmod N$, to occur, thereby giving us the desired discrete logarithm.

  In a very un-rigorous sense, forming a uniform superposition over N^2 *pairs* of elements could be thought of as allowing quantum interference to "detect collisions" within this space. In fact, upon measuring the ancilla register, the state will collapse to precisely a superposition of N pairs (a, b) which "collide" with the measured value, a *quadratic* reduction from N^2 elements down to N. This is pure philosophy however, so we best return to the mathematics.

Formulating the algorithm should feel like a routine at this point: apply $\mathcal{F}_{\mathbb{Z}_n \times \mathbb{Z}_n}$ to a primary register of $|0\rangle \otimes |0\rangle$, query the oracle U_f on an empty ancilla containing $|0\rangle$ (no need for a phase oracle in this case), measure the result of the ancilla, and then close the interference with $\mathcal{F}^\dagger_{\mathbb{Z}_n \times \mathbb{Z}_n}$ on the primary registers.

Using multiplicativity, we of course know that $\mathcal{F}_{\mathbb{Z}_n \times \mathbb{Z}_n} = \mathcal{F}_{\mathbb{Z}_n} \otimes \mathcal{F}_{\mathbb{Z}_n}$ and so we can execute the first QGFT via two ordinary QFTs, arriving at the uniform state;

$$\frac{1}{\sqrt{|\mathbb{Z}_n^2|}} \sum_{a=0}^{N-1} \sum_{b=0}^{N-1} |a\rangle |b\rangle |0\rangle \overset{U_f}{\rightsquigarrow} \frac{1}{N} \sum_{(a,b) \in \mathbb{Z}_n^2} |a\rangle |b\rangle \left|\alpha^a h^b\right\rangle$$

And hence upon measuring the ancilla we shall read out some random selection of an element $\alpha^a h^b \in A$, which is to say some α^m for a value of m unknown to us. Whatever it may be, the state post-observation collapses to those elements $(a, b) \in \mathbb{Z}_n^2$ such that $f(a, b) = \alpha^m = \alpha^m h^0 = f(m, 0)$: in other words, the state collapses to precisely a *coset* of our desired hidden subgroup $\ker f$:

$$\frac{1}{\sqrt{N}} \sum_{(a,b) \in (m,0) + \langle (n,-1) \rangle} |a\rangle |b\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |a = m + kn\rangle |b = -k\rangle$$

Take a moment to convince yourself of this intuitively: the elements of the superposition / group whose images are the observed value α^m are precisely those elements such that $\alpha^{a=m+kn} h^{b=-k} = \alpha^{m+kn}\alpha^{-kn} = \alpha^m$ as k cycles over $\mathbb{Z}_N$. What's so beautiful about the QGFT is then that the actual value of m ends up factoring through in an irrelevant way: allowing us to focus on the hidden *subgroup*

$\langle(n,-1)\rangle$ without the mask of the *coset* $(m,0)$. More precisely, upon applying two copies of $\mathcal{F}^{\dagger}_{\mathbb{Z}_n}$ we obtain;

$$\frac{1}{\sqrt{N}^2}\sum_{k_1=0}^{N-1}\sum_{k_2=0}^{N-1}|k_1\rangle|k_2\rangle\frac{1}{\sqrt{N}}\sum_{k=0}^{N-1}e^{2\pi ik_1/N[m+kn]}e^{2\pi ik_2/N[-k]}$$

$$=\frac{1}{\sqrt{N}}\sum_{(k_1,k_2)\in\mathbb{Z}_n^2}|k_1\rangle|k_2\rangle e^{2\pi ik_1/Nm}\frac{1}{N}\sum_{k\in\mathbb{Z}_n}\overline{e^{2\pi ik/N[k_2]}}e^{2\pi ik/N[k_1n]}=\frac{1}{\sqrt{N}}\sum_{(k_1,k_2)\in\mathbb{Z}_n^2:k_2\equiv nk_1}|k_1\rangle|k_2\rangle e^{i\square}$$

Where the familiar filtration identity reduces us down to a superposition over those states (or rather representations) such that $k_2\equiv k_1n \pmod N$, and where $e^{i\square}$ is meant to emphasize that any data concerning m is simply lost to a phase factor which has no influence on the measurement probability.

To repeat the $\mathcal{F}_G$ mantra one more time, notice how upon observing the primary registers, ie obtaining a pair (k_1,k_2), we have gained one *constraint* which the hidden subgroup H must satisfy: it must be that $n\equiv k_1^{-1}k_2 \pmod N$ which reveals the desired logarithm.

Theoretically, k_1 might not always be invertible modulo N, which is to say $d=\gcd(k_1,N)>1$, but even then, $k_1n\equiv k_2 \pmod N$ has at most d solutions which can be checked individually. If by some extraordinary luck d is still too large to refine classically[4], simply re-run the algorithm to produce an additional pair (k_1,k_2). Formally, it's not exactly hard to check that the probability of k_1 being coprime to N would be;

$$\frac{\varphi(N)}{N}=\frac{p_1^{e_1-1}(p_1-1)\dots p_k^{e_k-1}(p_k-1)}{p_1^{e_1}\dots p_k^{e_k}}=\left(1-\frac{1}{p_1}\right)\dots\left(1-\frac{1}{p_k}\right)$$

Which is tiny when the prime factors $p_1,\dots p_k$ of N are of any reasonable size, which is the only practically interesting case given that small prime factors in N can be easily removed via the delightful reduction method of Pohlig–Hellman.

Exercise

Skimming over the cryptography, when I visit `yale.edu`, a secure connection is established using Elliptic Curve Diffie-Hellman Key Exchange in a cyclic group of prime order $p\approx 2^{252}$ (see the `x25519` curve for more details). How many generators α does $\mathbb{Z}_p$ have? If you were to try the Birthday Attack described above, how much memory would you likely need (assume for simplicity that each stored value takes up 256 bits of memory)? Look up / estimate how many quantum gates would be required to implement $\mathcal{F}_{\mathbb{Z}_p}$ of this size: it now seems a lot more feasible to break that connection!

A Toy Qiskit Demonstration: S_3

Our goal here is simple: starting from nothing but the representations of S_3 (as in [1]), can we design our own quantum algorithm for determining the automorphism group of a graph on 3 vertices? To be clear, this section is *not* intended to communicate anything close to the legitimate quantum algorithm for the Graph Automorphism problem (for a real discussion of this fascinating task see [3]), but instead a check on our understanding of the QGFT.

To start, it's not hard to verify that the set of inequivalent, irreducible, unitary representations of the group S_3 are:

- The *trivial* representation $\mathbf{1}(g) = [1]$, of dimension 1
- The *sign* representation $\operatorname{sgn}(g) = [\operatorname{sgn}(g)] = [\pm 1]$, of dimension 1
- And the *standard representation* ρ: permutations of the standard basis vectors e_1, e_2, e_3 for $\mathbb{C}^3$ restricted to the 2-dimensional invariant subspace $V = \{x_1e_1 + x_2e_2 + x_3e_3 : x_1 + x_2 + x_3 = 0\}$.

Out of the three claimed above, the standard representation likely seems the least natural, but it's really not that strange: to each permutation in S_3 we can easily create a linear action on $\mathbb{C}^3$ by simply permuting the corresponding basis vectors e_1, e_2, e_3, for example $(23) \mapsto \left[\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right]$ which linearly swaps e_2 and e_3. However, this action is clearly not irreducible since V is trivially invariant, so we consider instead the restriction of the action of S_3 on $\mathbb{C}^3$ onto the 2-dimensional space V where it becomes irreducible.

In particular, notice that $1^2 + 1^2 + 2^2 = 6 = |S_3|$ gives us the appropriate count of dimensions3 for $\widehat{S_3}$, assuming you've checked they're all unitary, irreducilbe, inequivalent, etc.

Now, the 2×2 matrix characterization of ρ (recall, we want to extract the entries as Fourier modes $|\rho_{0,0}\rangle, |\rho_{0,1}\rangle, \dots$) come from picking a basis for V, typically $v_1 = \frac{1}{\sqrt{2}}(1, -1, 0)$ and $v_2 = \frac{1}{\sqrt{6}}(1, 1, -2)$ and observing how these are transformed under the action of $\sigma \in S_3$ on all of $\mathbb{C}^3$. For example, consider (12) acting on the point $\alpha v_1 + \beta v_2 = \left(\frac{\alpha}{\sqrt{2}} + \frac{\beta}{\sqrt{6}}, \frac{-\alpha}{\sqrt{2}} + \frac{\beta}{\sqrt{6}}, \frac{-2\beta}{\sqrt{6}}\right)$: we swap to obtain $\left(\frac{-\alpha}{\sqrt{2}} + \frac{\beta}{\sqrt{6}}, \frac{\alpha}{\sqrt{2}} + \frac{\beta}{\sqrt{6}}, \frac{-2\beta}{\sqrt{6}}\right)$ and then revert to our basis coordinates as $-\alpha v_1 + \beta v_2$. Thus, $\rho((12)) = \left[\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ since this is the action of the permutation (12) restricted to our chosen basis of v_1, v_2.

Exercise

Show that the standard representation given above really is irreducible, and sketch the subspace V if $\mathbb{C}$ were replaced with ordinary coordinates in $\mathbb{R}$. Where do the basis vectors v_1 and v_2 lie?

Show that $\rho((123)) = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$ with respect to the basis provided above. What *geometric* transformation is this matrix actually performing on the plane V? Does this match your intuition for what the cyclic group $\mathbb{Z}_3 \cong \langle (123) \rangle$ should do?

So, the first two columns of $\mathcal{F}_{S_3}$ are easy: the coefficient attached to $|\mathbf{1}\rangle$ is always $\frac{+1}{\sqrt{|G|}}$ and the coefficient attached to $|\operatorname{sgn}\rangle$ is always $\frac{+1}{\sqrt{|G|}}$ when the corresponding $g \in G$ has an even sign, as in $|e\rangle, |(123)\rangle, |(132)\rangle$ and $\frac{-1}{\sqrt{|G|}}$ when the corresponding element has odd sign, as in $|(12)\rangle, |(13)\rangle, |(23)\rangle$. Consequently;

$$\mathcal{F}_{S_3} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & 1 & \dots \\ 1 & 1 & \dots \\ 1 & 1 & \dots \\ 1 & -1 & \dots \\ 1 & -1 & \dots \\ 1 & -1 & \dots \end{bmatrix}$$

Now, the remaining 4 columns correspond to the matrix entries of $\rho(e), \rho((123)), \ldots \rho((23))$, which we can compute using homomorphism rules. For example, since $\rho((123)) = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$, it must be that $\rho((132)) = \rho((123))^2 = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$ and similarly since $(23) = (12) \circ (123)$ we see that $\rho((23)) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$. Indeed, since S_3 is generated by (12) and (123) we can form all of the desired matrices for all of the remaining elements using the two we've found already and extract the 4 entries $\rho_{0,0}, \rho_{0,1}, \rho_{1,0}, \rho_{1,1}$ within each one, being sure not to forget the $\sqrt{d_\rho} = \sqrt{2}$ scale factor. In total;

$$\mathcal{F}_{S_3} = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{1}{2\sqrt{3}} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{1}{2\sqrt{3}} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} & 0 & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{2\sqrt{3}} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{2\sqrt{3}} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2\sqrt{3}} \end{bmatrix}$$

Which we can check is indeed a unitary matrix! In terms of Qiskit/Python code, our implementation currently stands as;

```python
import numpy as np
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
import json

QGFT=np.array([
    [1/np.sqrt(6),  1/np.sqrt(6),  1/np.sqrt(3),       0,     0,  1/np.sqrt(3)],
    [1/np.sqrt(6),  1/np.sqrt(6), -1/(2*np.sqrt(3)), -0.5,   0.5, -1/(2*np.sqrt(3))],
    [1/np.sqrt(6),  1/np.sqrt(6), -1/(2*np.sqrt(3)),  0.5,  -0.5, -1/(2*np.sqrt(3))],
    [1/np.sqrt(6), -1/np.sqrt(6), -1/np.sqrt(3),       0,     0,  1/np.sqrt(3)],
    [1/np.sqrt(6), -1/np.sqrt(6),  1/(2*np.sqrt(3)), -0.5,  -0.5, -1/(2*np.sqrt(3))],
    [1/np.sqrt(6), -1/np.sqrt(6),  1/(2*np.sqrt(3)),  0.5,   0.5, -1/(2*np.sqrt(3))]
]) #The QGFT matrix we compued

QGFT_8=np.block([
    [QGFT,np.zeros((6,2))],
    [np.zeros((2,6)),np.eye(2)]
]) #3 registers = 8 logical states -> 2 dummy / leftover states.

print("Should be Hermitian: ", QGFT.T @ QGFT)

qc = QuantumCircuit(6,6) #We'll need 3 qubits and 3 ancillas, together with space to measure each.

qc.unitary(QGFT_8, [0, 1,2]) #An arbitrary unitary gets decomposed into a product of known gates.

qc.measure([0, 1, 2], [0, 1, 2])
result = AerSimulator().run(qc, shots=1024).result()
print(json.dumps(result.get_counts(),sort_keys=True))
```

Where importantly our 6 group element states $|e\rangle, |(123)\rangle, \ldots$ have been encoded as the binary strings $000, 001, \ldots$ with the remaining 7th and 8th states being dummy placeholders. As expected, $\mathcal{F}_{S_3}(|000\rangle)$ produces a uniform superposition over each of the 6 S_3 states, as we can confirm with the printed tally.

Of course, to extract the automorphism group of an object we need some way for the environing group G to act upon it. In our case we will manually (and extremely inefficiently) construct such an oracle interface by applying each $\sigma \in S_3$ to a graph defined by a list of `Edges`, then search through a database to see which of the $2^{\binom{3}{2}} = 8$ possible graph configurations the resulting image landed in. Needless to say, enumerating over all $n!$ permutations to manually encode exactly this behavior into a unitary gate is not practical, but what matters for our purposes is that such an oracle theoretically exists.

```python
def GraphOracle(i):
    import itertools
    S3=[
```

```python
        [1,2,3], #Identity
        [2,3,1], #3 cycle
        [3,1,2], #other 3 cycle
        [2,1,3], #Swap 1 and 2
        [3,2,1], #Swap 1 and 3
        [1,3,2]  #Swap 2 and 3
    ]
    Edges=[[1,2],[2,3]] #For example, a line graph: 1-2-3

    sigma=S3[i]
    NewEdges=[[sigma[vertex-1] for vertex in edge] for edge in Edges] #Apply sigma to the graph

    possible_edges = list(itertools.combinations([1,2,3], 2))

    #There are 2^{3 choose 2} possible graphs on 3 vertices: let's enumerate all of them.
    AllGraphs = []
    for r in range(len(possible_edges) + 1):
        for subset in itertools.combinations(possible_edges, r):
            AllGraphs.append([list(edge) for edge in subset])

    #One of the possible graphs will match our NewEdges, we return which one it corresponds to.
    for i,Graph in enumerate(AllGraphs):
        if sorted([sorted(edge) for edge in NewEdges])==sorted([sorted(edge) for edge in Graph]):
            break
    return i

OracleDict={i:GraphOracle(i) for i in range(6)}

def QiskitShouldHaveThisBuiltInButItDoesntForSomeReason(dict,in_bits,out_bits):
    from qiskit import QuantumCircuit, QuantumRegister
    from qiskit.circuit.library import MCXGate

    '''
    To emulate what a real oracle would be doing,
    we assign a Multi-Controlled-Not gate (MCX) to each possible input x,
    assigned the task of flipping one bit of the target register
    when the input register matches the desired state x.
    '''

    PrimaryRegister=QuantumRegister(in_bits, 'input')
    AncillaRegister=QuantumRegister(out_bits, 'target')
    qc = QuantumCircuit(PrimaryRegister,AncillaRegister)

    for x in dict.keys():
        y = dict[x]

        # Sneaky python trick to turn integers into Qiskit bit ordering.
        input_bin = format(x, '03b')[::-1]
        output_bin = format(y, '04b')[::-1]

        # Suppose you wanted the MCX gate to fire on the state 101.
        # You would need to X the 2nd qubit so that all three are in the +1 state.
        for i, bit in enumerate(input_bin):
            if bit == '0':
                qc.x(PrimaryRegister[i])

        # Apply the MCX gate for every 1 bit outputted by the function oracle
        for j, bit in enumerate(output_bin):
            if bit == '1':
                qc.append(MCXGate(in_bits), [*PrimaryRegister, AncillaRegister[j]])

        # Then we must undo the X gates added in the first step (uncompute).
        for i, bit in enumerate(input_bin):
            if bit == '0':
                qc.x(PrimaryRegister[i])
    return qc.to_gate()

Uf=QiskitShouldHaveThisBuiltInButItDoesntForSomeReason(OracleDict,3,3)
```

At this point there should be no hesitation as to what the sampling circuit looks like: $\mathcal{F}_G U_f \mathcal{F}_G^\dagger$

```python
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator, Aer
```

```python
qc = QuantumCircuit(6,6) #We will need 3 qubits and 3 ancillas, together with space to measure each one.

qc.unitary(QGFT_8, [0, 1,2]) #An arbitrary unitary gets decomposed into a product of known gates.

qc.append(Uf,range(6)) #Apply the oracle gate we brute-forced into existence.

qc.measure([3,4,5],[3,4,5]) #Measure the ancilla bits: collapsing into a coset state.

qc.unitary(np.transpose(QGFT_8),[0,1,2]) #Close the interference with the QGFT:

qc.measure([0, 1, 2], [0, 1, 2]) #We will now read what representation we measured!

#Boilerplate code to run Qiskit:
backend = Aer.get_backend('qasm_simulator')
result = backend.run(transpile(qc, backend), shots=1024).result()
counts = result.get_counts()
```

Now, to make sense of the resulting measurements recall from our initial construction of the QGFT that $\mathcal{F}_G$ is well-suited to distinguish between subgroups $H \leq G$ via their corresponding *projection* operator $\Pi_\rho^H = \frac{1}{|H|}\sum_{h\in H}\rho(h)$, say from 5 and 6. In particular, recall that the entries of `result.get_counts()` will consist of the number of measurements we have made of the $|1\rangle$ representation in $\left|\widehat{G}\right\rangle$, the number of measurements we made of the $|\text{sgn}\rangle$ representation, and finally the 4 entries in $|\rho_{0,0}\rangle, |\rho_{0,1}\rangle, \ldots$. Let's work through some examples to get a feel for these two objects:

- For any subgroup H, it's trivial that $\Pi_1^H = \frac{1}{|H|}\sum_{h\in H}[1] = [+1]$.
- S_3 has 3 subgroups isomorphic to $\mathbb{Z}_2$, namely those generated by the two-cycles $(12), (13), (23)$. Each contains one odd and one even permutation so $\Pi_{\text{sgn}}^{\mathbb{Z}_2} = [0]$ averages out to 0, and the same balance of even-and-odd applies to $\Pi_{\text{sgn}}^{S_3} = [0]$ as well.

  This cancellation tells us that if the hidden subgroup H has an even size, then there is no amount of the Fourier mode $|\text{sgn}\rangle$ "present" in H: there is nothing to "project" onto V_ρ^H.
- There is a unique copy of $\mathbb{Z}_3$ in S_3 generated by (123) and hence $\Pi_\rho^{\mathbb{Z}_3} = \frac{1}{\sqrt{3}}\rho^0((123)) + \frac{1}{\sqrt{3}}\rho^1((123)) + \frac{1}{\sqrt{3}}\rho^2((123))$ which we can compute explicitly using $\rho((123)) = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$ to find that $\Pi_\rho^{\mathbb{Z}_3} = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ also vanishes as well.

  Similarly, since there is no "component" of $\mathbb{Z}_3$ "inside" of ρ, if the hidden subgroup is ever $H = \mathbb{Z}_3$, we will have a 0% chance of extracting any of the Fourier modes $|\rho_{0,0}\rangle, |\rho_{0,1}\rangle, \ldots$ from it.

And we leave it to the determined reader to complete the rest of the entries in the table, recalling that the probability $P(\rho)$ of measuring a representation ρ from the output of the QGFT is proportional to $d_\rho \text{tr}(\Pi_\rho^H)$ 6:

H	Π_1^H	Π_{sgn}^H	Π_ρ^H	$P(1)$	$P(\text{sgn})$	$P(\rho)$
1	$[1]$	$[1]$	$\left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$	0.167	0.167	0.667
$\mathbb{Z}_3$	$[1]$	$[1]$	$\left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right]$	0.50	0.50	0.00
$\mathbb{Z}_2 \cong \langle(12)\rangle$	$[1]$	$[0]$	$\frac{1}{2}\left[\begin{smallmatrix} 0 & 0 \\ 0 & 2 \end{smallmatrix}\right]$	0.333	0.00	0.667
$\mathbb{Z}_2 \cong \langle(13)\rangle$	$[1]$	$[0]$	$\frac{1}{2}\begin{bmatrix} \frac{3}{2} & \frac{-\sqrt{3}}{2} \\ \frac{-\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$	0.333	0.00	0.667
$\mathbb{Z}_2 \cong \langle(23)\rangle$	$[1]$	$[0]$	$\frac{1}{2}\begin{bmatrix} \frac{3}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$	0.333	0.00	0.667
S_3	$[1]$	$[0]$	$[0]$	1.00	0.00	0.00

Table 1: Possible hidden subgroups H, their corresponding projection operators Π^H, and the resulting weakly-sampled distribution P.

Notice in particular that the probability distributions for each copy of $\mathbb{Z}_2$ are completely indistinguishable![2] The QGFT on its own has successfully discriminated between the subgroups of S_3, but only up to isomorphism, hence why in the study of the non-Abelian HSP, care is taken to characterize what data can be extracted from weak vs. strong Fourier sampling **??**.

Here is some helper code to tidy up the Qiskit counts into a distribution matching the above table:

```python
# We don't care about the ancilla measurements, so pool all the results together by the first 3 bits (primary register).
aggregated = {state: 0 for state in [format(i, '03b') for i in range(8)]}
for bitstring, count in counts.items():
    input_bits = bitstring[-3:]
    if input_bits in aggregated:
        aggregated[input_bits] += count

# Distibution over which representations are observed
# (recall that for our 2-dim representation, we pool together all 4 entries).
FinalDistribution=[aggregated['000'],aggregated['001'],
aggregated['010']+aggregated['011']+aggregated['100']+aggregated['101']]
print(np.array(FinalDistribution)/np.sum(FinalDistribution))
```

And indeed: When we set `Edges=[[1,2],[2,3]]`, a line graph having automorphism group of $\mathbb{Z}_2 \cong \langle(13)\rangle$ formed from swapping symmetrically the endpoints, the counts come back as `[0.3251, 0., 0.6749]` as predicted! When we try a fully symmetric triangle graph with `Edges=[[1,2],[2,3],[3,1]]`, the distribution reads as `[1., 0., 0.]` too. The same holds true for graphs like `Edges=[[1,3]]` which also have automorphism group $\mathbb{Z}_2$ and the empty graph `Edges=[]` which is perfectly symmetric. Unfortunately, it's pretty trivial to see that there is no graph on 3 vertices with automorphism group $H = 1$ or $H = \mathbb{Z}_3$ so we cannot actually test these cases, but we nonetheless conclude that the observed values match the theoretical predictions.

Conclusion

The remarkable interplay between group representations and the quantum algorithms which can be synthesized from them was not an overnight discovery:[2] the pioneering 1994 algorithms of Simon and Shor came chronologically first, followed by incremental improvements into the understanding of Abelian stabilizers by Kiyatev, Boneh, and Lipton in 1995, with the popularization of the phrase "Hidden Subgroup Problem" appearing some time after the wave of theoretical advancements in quantum algorithms of the late 1990's[6]. In fact, only as recently as 2004 was it established by Ettinger, Hoyer, and Knill[7] that QGFT methods could even extract a sufficient amount of "data" (with a polynomial-time query complexity of course) from an HSP oracle on a general group: let alone process it in under exponential time. Akin to Riemannian geometry and Einstein's General Relativity or Boole's logical calculus and the modern programmable computer, it is immensely inspiring to see how ostensibly self-contained, abstract pieces of mathematics (such as representation theory for example) end up speaking an almost *eerily perfect* language necessary to formalize cutting-edge advances in the physical sciences.

In addition to encouraging the reader to explore the applied and theoretical details of the HSP further, we would also like to close by mentioning that the divide between the Abelian QGFT and the non-Abelian QGFT is *not* a "discontinuous shelf" in query complexity. Even within the "least Abelian" group possible, S_n, efficient algorithms are known for the polynomial-time reconstruction of at least the *normal core* $\bigcap_{g\in G} gHg^{-1}$ of a hidden subgroup H for example[3]. Groups such as $D_{2n} = \langle r, s : r^n = s^2 = srsr = 1\rangle$, ie the *Dihedral Group* of rigid symmetries of a regular n-gon, are in some sense "just barely Abelian" owing to the very simple commutation relation $srs^{-1} = r^{-1}$, and as a result, the Dihedral QGFT[2] and the Dihedral HSP are some of the most actively studied sub-problems of the genre in the present day. Only time will tell where further developments in group-theoretic Fourier sampling and its hardware implementations may lead.

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